

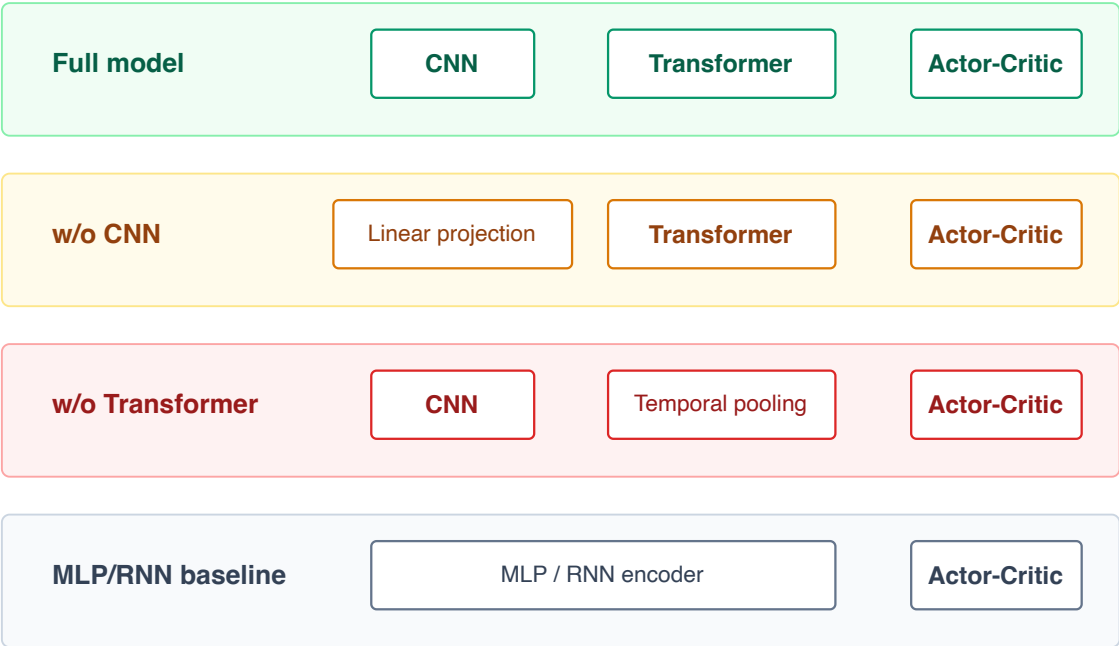
Ablation Study of CNN and Transformer Modules

One-figure protocol for attributing local waveform modeling and long-range temporal modeling in physiological VAD control

A Unified Evaluation Protocol



B Encoder Variants for Module Attribution



C Contribution Attribution Matrix

Variant	Perfusion matching	Transition adaptation	Unsafe-event avoidance	Control smoothness
Full	Best	Best	Best	Best
w/o CNN	Drop	Mid	Drop	Mid
w/o TF	Mid	Drop	Drop	Drop
Baseline	Low	Low	Low	Low

Interpretation: performance rank is computed after replacing these categorical labels with measured normalized metrics.

D Module Contribution Takeaways

CNN: captures local waveform morphology and short-term unsafe patterns

Transformer: models cross-cycle dependencies and physiological transitions

Full encoder: best safety-performance tradeoff for adaptive VAD speed regulation